Close

```
Name : Vaibhav Patil
Roll No : DW234
Project Title : Comphrensive Overview of W
100 Univesities
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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv("A Comprehensive Overview of the Worlds Top 1000 Universities a
```

```
import pandas as pd

df = pd.read_csv("data.csv")
df.head()
```

	World Rank	Institution	Location	National Rank	Quality of Education	Alumni Employment	Quality of Faculty	Rese Ou
0	1	Harvard University	USA	1	2.0	1	1.0	
1	2	Stanford University	USA	2	10.0	3	2.0	
2	3	Massachusetts Institute of Technology	USA	3	3.0	11	3.0	
3	4	University of Cambridge	United Kingdom	1	5.0	19	6.0	
4	5	University of Oxford	United Kingdom	2	9.0	25	10.0	

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(1000, 14)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   World Rank            1000 non-null   int64
 1   Institution           1000 non-null   object
 2   Location              1000 non-null   object
 3   National Rank         1000 non-null   int64
 4   Quality of Education  1000 non-null   float64
 5   Alumni Employment     1000 non-null   int64
 6   Quality of Faculty    1000 non-null   float64
 7   Research Output       1000 non-null   int64
 8   Quality Publications  1000 non-null   int64
 9   Influence             1000 non-null   int64
 10  Citations             1000 non-null   int64
 11  Score                 1000 non-null   float64
 12  Latitude              1000 non-null   float64
 13  Longitude             1000 non-null   float64
dtypes: float64(5), int64(7), object(2)
memory usage: 109.5+ KB
```

```
df.describe()
```

	World Rank	National Rank	Quality of Education	Alumni Employment	Quality of Faculty	Research Output
count	1000.000000	1000.00000	1000.000000	1000.000000	1000.000000	1000.000000
mean	500.500000	40.02300	528.532000	726.528000	261.180000	508.454000
std	288.819436	49.22871	208.781322	343.686144	84.233419	300.080433
min	1.000000	1.00000	1.000000	1.000000	1.000000	1.000000
25%	250.750000	6.00000	405.750000	425.750000	282.750000	250.750000
50%	500.500000	21.00000	667.000000	1001.000000	304.000000	500.500000
75%	750.250000	51.00000	667.000000	1001.000000	304.000000	754.250000
max	1000.000000	213.00000	667.000000	1001.000000	304.000000	1001.000000

The dataset contains information about the top 1000 universities including ranking, country, and performance scores.

```
df.isnull().sum()
```

	0
World Rank	0
Institution	0
Location	0
National Rank	0
Quality of Education	0
Alumni Employment	0
Quality of Faculty	0
Research Output	0
Quality Publications	0
Influence	0
Citations	0
Score	0
Latitude	0
Longitude	0

dtype: int64

No missing values found in the dataset.

```
df.drop_duplicates(inplace=True)
```

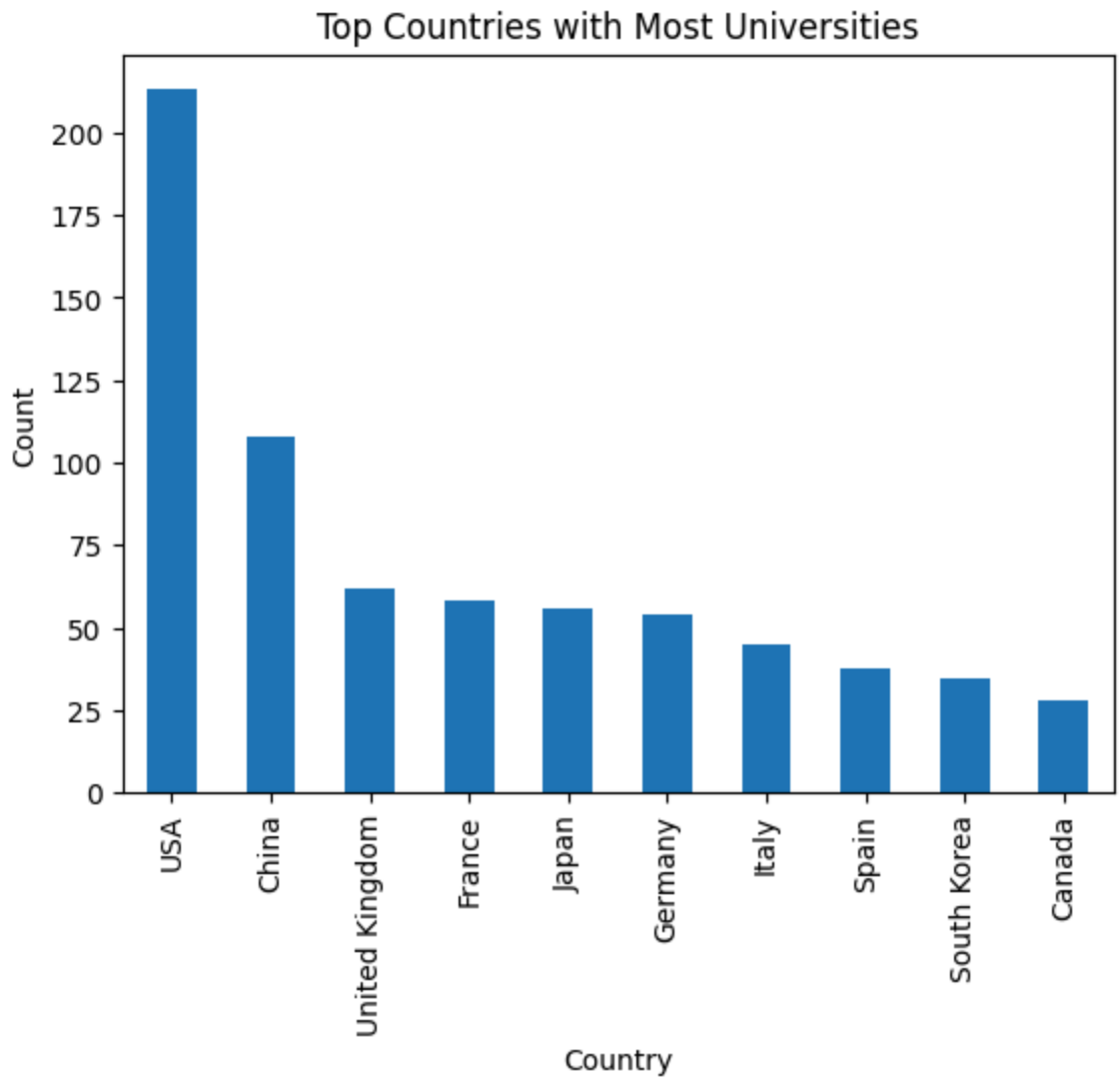
Duplicates were removed and missing values were handled.

```
df.sort_values(by='Score').head(10)
```

	World Rank	Institution	Location	National Rank	Quality of Education	Alumni Employment	Quality of Faculty	R
999	1000	Capital Normal University	China	108	667.0	869	304.0	
998	999	Soonchunhyang University	South Korea	35	667.0	1001	304.0	
997	998	K?chi University	Japan	56	667.0	1001	304.0	
996	997	University of Calcutta	India	17	353.0	716	296.0	
993	994	Loma Linda University	USA	213	180.0	1001	304.0	
987	988	Manchester Metropolitan University	United Kingdom	62	667.0	960	304.0	
995	996	Aga Khan University	Pakistan	3	667.0	1001	304.0	
994	995	University of Macau	Macau	1	667.0	512	304.0	
989	990	Khon Kaen University	Thailand	4	667.0	1001	304.0	
990	991	University of Ibadan	Nigeria	1	540.0	1001	265.0	

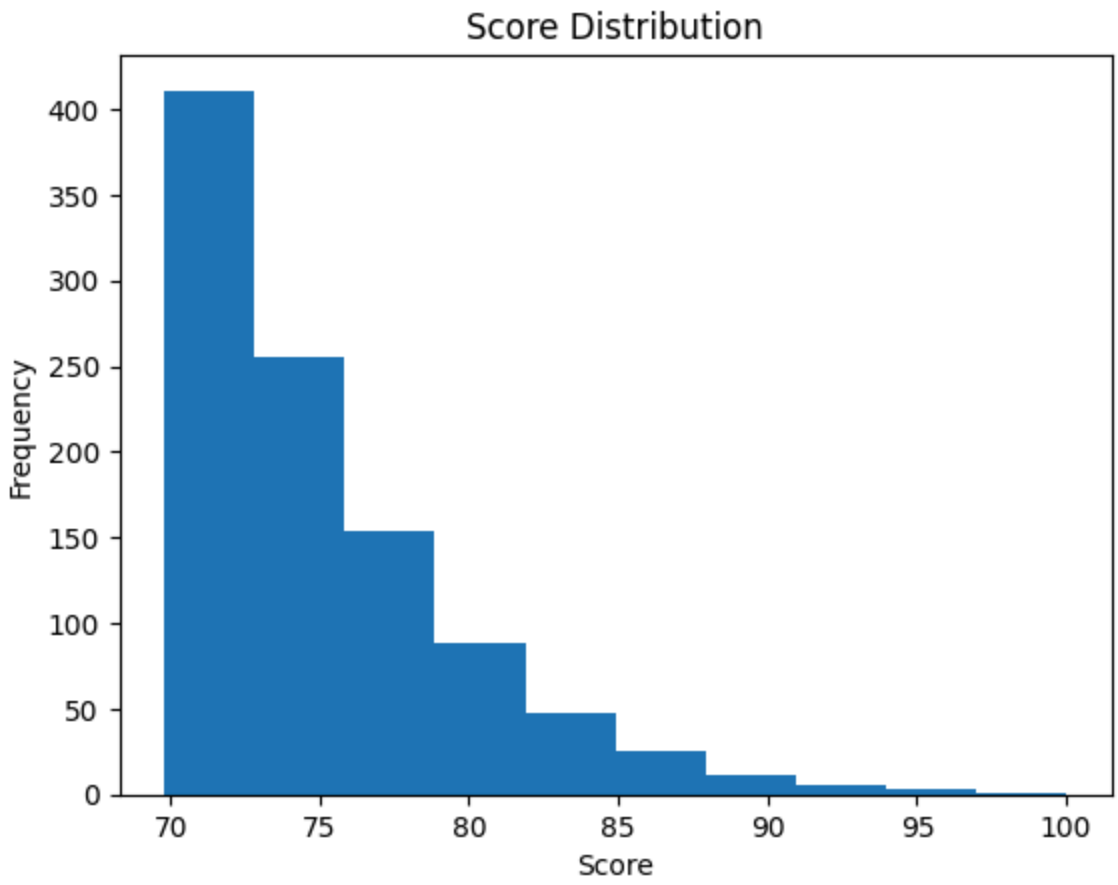
Top universities are ranked based on performance indicators.

```
df['Location'].value_counts().head(10).plot(kind='bar')
plt.title("Top Countries with Most Universities")
plt.xlabel("Country")
plt.ylabel("Count")
plt.show()
```



Some countries have more universities in the top rankings.

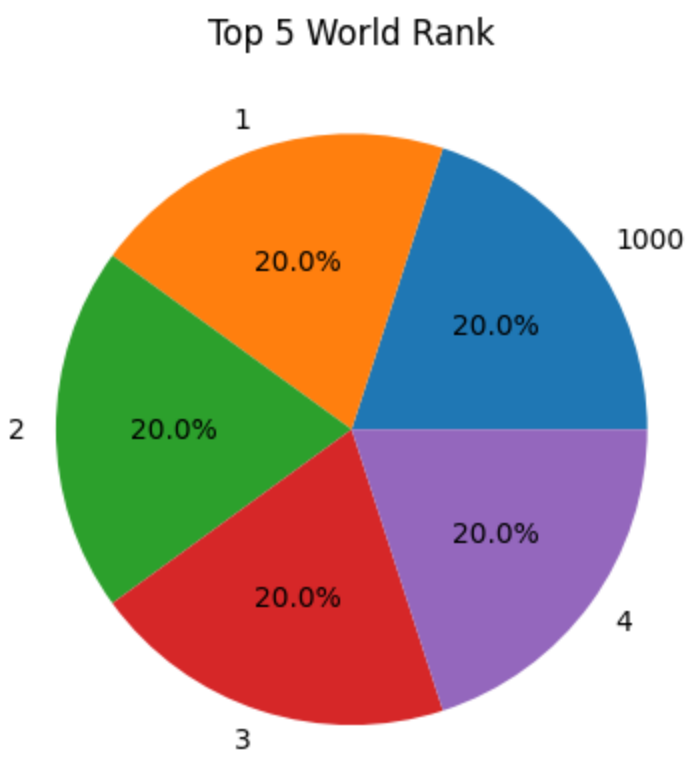
```
plt.hist(df['Score'])
plt.title("Score Distribution")
plt.xlabel("Score")
plt.ylabel("Frequency")
plt.show()
```



Most universities have mid-range scores.

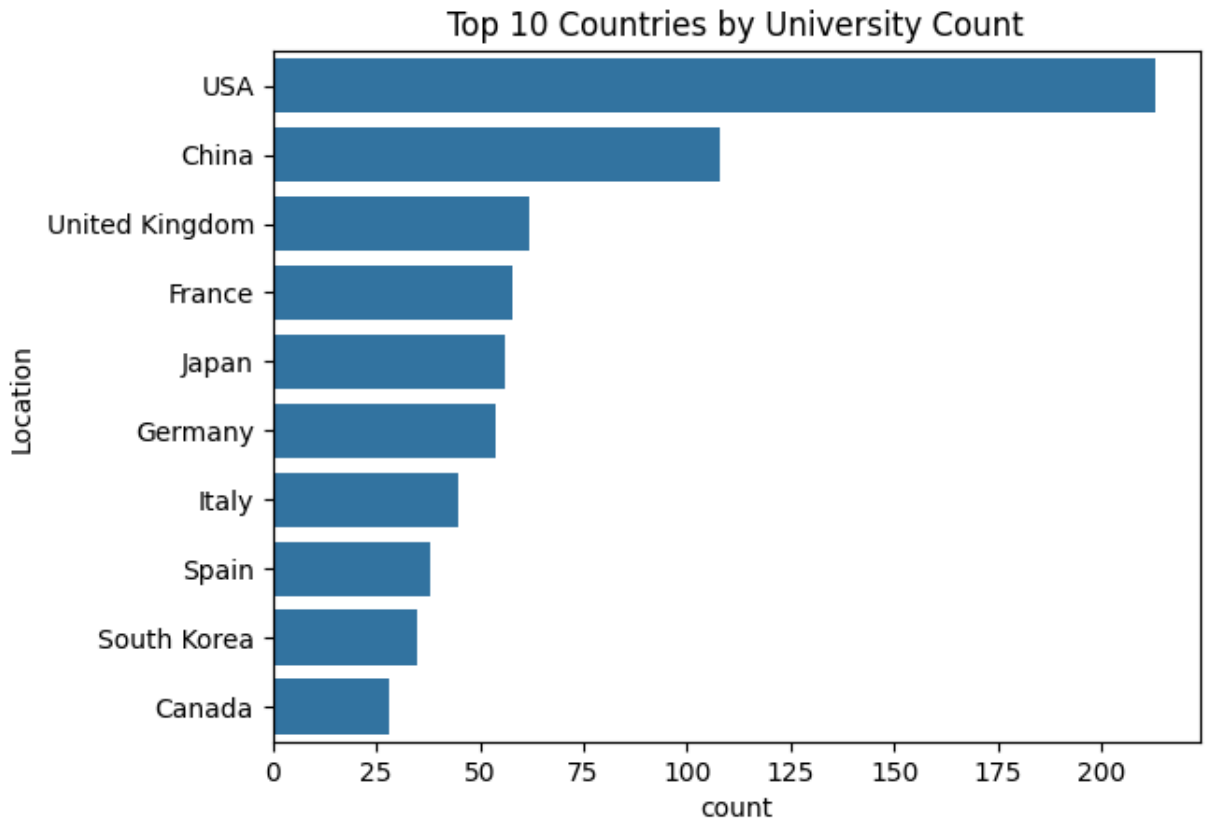
```
top_countries = df['World Rank'].value_counts().head(5)

plt.pie(top_countries, labels=top_countries.index, autopct='%1.1f%%')
plt.title("Top 5 World Rank")
plt.show()
```



Top countries contribute the majority of universities in the dataset.

```
sns.countplot(y='Location', data=df, order=df['Location'].value_counts().index[
plt.title("Top 10 Countries by University Count")
plt.show()
```



Some countries dominate global university rankings.

```
plt.scatter(df['Score'], df['World Rank'])
plt.title("Score vs World Rank")
plt.xlabel("Score")
plt.ylabel("Rank")
plt.show()
```

Score vs World Rank